

Software Design Diagram Generator

Product Backlog / User Stories

IT314 Software Engineering · Group 27

155 user stories · 13 EPICs · 4 sprints

Derived from Requirements

How to read this document

- **Front of card** = the user story (role / goal / reason). **Back of card** = Success Criteria (✓) and Failure / Edge Criteria (X). Every criterion is testable.
- Each story carries its EPIC and the sprint it is planned for (see the EPICs & Sprint Plan document). Story IDs are the SRS requirement ID plus a letter, for example FR3c is the third story of FR3.

Stories per sprint

Sprint	Stories
Sprint 1	33
Sprint 2	41
Sprint 3	42
Sprint 4	39
Total	155

1. Functional Requirement Stories

FR1 — Input Methods

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR1a	As a user, I want to type or paste requirement text directly, so I can start without a file.	✓ Text is saved as an input source in the current project and appears in the project's source list. ✓ Text is split into numbered requirement statements (FR4).	- X Empty or whitespace-only submission is rejected with a message; nothing is saved. - X Pasted text counts toward the 100-page project limit at 3,000 characters per page (rounded up); text that would exceed the limit is blocked (FR3i).	E-B Sprint 1
FR1b	As a user, I want to upload a PDF SRS document, so I can reuse a existing document.	✓ A .pdf of at most 10 MB is accepted, stored, and queued for extraction; its page count is recorded.	X File type is verified by content, not by extension; a non-PDF renamed to .pdf is rejected with a "not a valid PDF" message.	E-B Sprint 2
FR1c	As a user, I want to upload a DOCX SRS document, so I can reuse a existing document.	✓ A .docx of at most 10 MB is accepted, stored, and queued for extraction; its page count is recorded.	- X A password-protected .docx is rejected with a specific "protected file" error. - X File type is verified by content, not by extension.	E-B Sprint 2
FR1d	As a user, I want to input existing user stories directly, so I don't have to convert them to prose first.	✓ Stories in the "As a ... I want ... so that ..." pattern are parsed into role / goal / reason fields. ✓ Each story becomes one requirement statement.	X A malformed story (for example, missing "so that") is still accepted as a free-text requirement statement and marked "unstructured"; it is never silently dropped.	E-B Sprint 3
FR1e	As a user, I want to input a existing FR/NFR list directly, so I can reuse prior requirements work.	✓ A numbered FR/NFR list is parsed with its original IDs (for example "FR3") preserved as labels on the requirement statements.	X An unnumbered list is ingested as free text and the user is told that IDs could not be preserved.	E-B Sprint 3

FR2 — Reject Invalid Input

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR2a	As a user, I want to be told when my file exceeds 10 MB, so I know to shrink or split it.	✓ A file larger than 10 MB (10 × 1,048,576 bytes) is blocked with a message showing its actual size and the limit. ✓ The check runs in the browser before transfer and again on the server. ✓ No extraction or diagram generation runs on the rejected file.	X A file of exactly 10 MB is accepted (no off-by-one rejection).	E-B Sprint 2

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR2b	As a user, I want to be told when my file type is unsupported or corrupted, so I know it wasn't silently ignored.	✓ An unsupported type (for example .txt, .rtf, .png) is rejected with a message naming the supported types (PDF, DOCX). ✓ No extraction or diagram generation runs on the rejected file.	X A corrupted PDF or DOCX (fails to open) gives a distinct "file appears corrupted" message, not a generic failure.	E-B Sprint 2
FR2c	As a user, I want to be told when a PDF has no extractable text, or only some pages do, so I understand why nothing, or only part, was extracted.	✓ A PDF in which no page has a text layer (scanned/image-only) is rejected before extraction with a message that scanned PDFs are not supported. ✓ A PDF with both text and image-only pages is accepted; text pages are extracted and a warning lists each skipped page number. ✓ No requirement statement is ever generated from a skipped page.	- X Skipped pages still count toward the 100-page project limit. - X The warning stays visible on the document in the project's document list until the document is deleted.	E-B Sprint 2

FR3 — Project Management

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR3a	As a user, I want to create a new project with a name, so I can organize a set of requirements.	✓ The named project appears in my project list immediately and I become its Owner. ✓ Names are 1–100 characters, trimmed of leading and trailing spaces.	- X An empty name is rejected. - X A name that duplicates (case-insensitive) another project I own is rejected with a message; duplicates are not allowed.	E-B Sprint 1
FR3b	As a project owner, I want to rename a existing project, so I can keep names accurate.	✓ The new name is reflected in the project list, the project header, and generated reports. ✓ Only Owners can rename.	- X An empty or duplicate name is rejected and the old name is kept. - X Renaming does not change the project ID or any links.	E-B Sprint 2
FR3c	As a user, I want to see a list of all my projects, so I can find the one I need.	✓ The list shows every project I own and every project shared with me (shared projects are labelled "Shared", per FR24a), sorted by last-modified date, newest first.	X With no projects, a empty state with a "Create your first project" prompt is shown, not a blank screen.	E-B Sprint 1
FR3d	As a user, I want to switch between projects, so I can work on multiple efforts without losing data.	✓ Switching loads that project's requirements, extraction state, and diagrams completely.	- X Unsaved diagram edits trigger a Save / Discard / Cancel prompt before switching. - X Extracted-item edits are saved immediately and never trigger the prompt.	E-B Sprint 2

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR3e	As a project owner, I want to delete a project, so I can remove work I no longer need.	✓ Deletion requires a confirmation dialog in which I type the project name. ✓ Deletion permanently removes the project and all its documents, diagrams, versions, flags, feedback, and activity log. ✓ Only Owners can delete; collaborators are notified in-app.	X A double-click does not trigger two delete calls; deleting a already-deleted project shows “project not found”, not a error.	E-B Sprint 2
FR3f	As a user, I want to upload multiple documents into one project, so I can combine related sources.	✓ Multiple files attach to the same project and appear in a document list with name, page count, and status.	X If file 3 of 5 fails, files 1–2 stay uploaded and file 3 is listed with its failure reason.	E-B Sprint 2
FR3g	As a user, I want to combine text, uploaded files, user stories, and FR/NFR lists within one project, so I'm not limited to one input type.	✓ All input types from FR1a–e can coexist in one project and all are included in extraction.	X Conflicting information between two sources (for example, the same actor named differently) is flagged as an inconsistency (FR6c), not silently merged.	E-B Sprint 3
FR3h	As a system/admin, I want to block user from uploading beyond 15 documents per project, so the project stays manageable.	✓ Only uploaded files count as documents; the 16th upload attempt is blocked with a “15 document limit reached” message. ✓ Typed/pasted text, user-story input, and FR/NFR-list input do not count toward the 15-document limit (they count toward the page limit).	X Deleting a document frees a slot and uploads can proceed.	E-B Sprint 2
FR3i	As a system/admin, I want to block user from uploading beyond 100 total pages per project, so the project stays within processing limits.	✓ An upload or paste that would push the project total above 100 pages is blocked before processing, showing current and limit page counts. ✓ Pages are counted as: PDF = actual pages; DOCX = pages when rendered; pasted text = 1 page per 3,000 characters (rounded up).	- X A single file that alone exceeds 100 pages is rejected outright, not partially processed. - X Deleting a document frees its pages.	E-B Sprint 2

FR4 — Requirement Segmentation & Extraction

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR4g	As a user, I want my input split into individually identified requirement statements, so flags, traceability, and coverage counts can refer to one specific requirement.	✓ Every statement gets a ID unique within the project (R-001, R-002, ...) and shows its source document and location (page/paragraph). ✓ Original labels such as “FR3” or a user-story title are preserved as the statement's label.	- X A passage that cannot be split confidently is kept as one statement. - X IDs are never reused after a statement is deleted.	E-C Sprint 1

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR4a	As a user, I want the system to extract actors from my input, so I don't have to identify them manually.	✓ The actors list is populated after analysis, and each actor links to its source requirement statement(s).	X If no actors are found, a empty list is shown with a explanatory note, not a silent failure.	E-C Sprint 1
FR4b	As a user, I want the system to extract entities/classes from my input, so class-level modelling information is ready.	✓ The entities/classes list is populated with attributes where the input states them, each linked to its source statement(s).	X An entity that is mentioned but never defined still appears, marked "minimal".	E-C Sprint 1
FR4c	As a user, I want the system to extract processes from my input, so activity and DFD modelling information is ready.	✓ The processes list is populated and each process is linked to its source statement(s).	X A verb phrase that may not be a process is included but marked low-confidence for review, not presented as fact.	E-C Sprint 1
FR4d	As a user, I want the system to extract data flows and data stores from my input, so DFDs can be generated accurately.	✓ Data flows are listed as source → target with a description of the data; sources and targets can be actors, processes, or data stores. ✓ Data stores named or implied in the input are listed separately.	X A data flow with no clear destination is kept and flagged incomplete, not dropped.	E-C Sprint 1
FR4e	As a user, I want the system to extract interactions between actors and entities, so sequence diagrams have what they need.	✓ The interactions list is populated with ordering information where it can be inferred.	X An interaction whose order cannot be determined from the text is marked "order uncertain".	E-C Sprint 1
FR4f	As a user, I want the system to extract relationships between elements, so class and use-case diagrams show correct connections.	✓ The relationships list shows source, target, and type (association, inheritance, aggregation, composition, dependency, include, extend).	X A relationship whose type cannot be inferred defaults to "association" and is flagged for review.	E-C Sprint 1

FR5 — Review, Edit & Confirm Extracted Information

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR5a	As a user, I want to view all extracted information in one place before generation, so I can check it is correct.	✓ One view lists all FR4 categories (actors, entities, processes, data flows, data stores, interactions, relationships) grouped by category.	X With more than 200 items, each category is paginated at 50 items per page, not shown as one wall of text.	E-C Sprint 1
FR5b	As a user, I want to edit a extracted item, so I can correct a misread actor, entity, or process.	✓ The edited value is saved immediately, marked "user-edited", and reflected in any downstream diagram preview.	X Editing to a empty value is rejected and the prior value is kept.	E-C Sprint 1
FR5c	As a user, I want to add a missing extracted item, so nothing the system missed is left out.	✓ The added item appears in its category, tagged "user-added" (as opposed to "system-extracted").	X An item with the same type and name (case-insensitive) as a existing one triggers a "Keep both / Cancel" prompt, not a silent duplicate.	E-C Sprint 2
FR5d	As a user, I want to delete a incorrectly extracted item, so it doesn't appear in generated diagrams.	✓ The deleted item is removed from the list and excluded from all diagrams generated afterwards.	X Deleting a item that other items depend on (for example, a actor used in a interaction) shows a warning listing the dependents before confirming.	E-C Sprint 2

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR5e	As a user, I want to explicitly confirm the extracted information before diagrams are generated, so nothing proceeds without my sign-off.	✓ The Confirm action gates the transition to diagram-type selection and generation; the Generate button is disabled until the current extraction is confirmed.	X Any change to extracted items after confirmation clears the confirmation; I must confirm again. X Skipping confirmation (for example, by direct URL or API call) is rejected on the server, not just hidden in the UI.	E-C Sprint 1

FR6 — Detect Issue Types

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR6a	As a user, I want ambiguous requirements to be flagged, so I can clarify them before they affect diagrams.	✓ Ambiguous phrasing (for example “appropriate viewers”, “fast”, “user-friendly”) is detected and a flag of type Ambiguous is attached to the requirement statement.	X A requirement with no ambiguity indicators is not flagged (no false positive on clean input).	E-C Sprint 2
FR6b	As a user, I want incomplete requirements to be flagged, so I know where information is missing.	✓ A requirement missing key information (for example, no actor named, no measurable value) gets a flag of type Incomplete.	X A terse but complete requirement is not flagged.	E-C Sprint 2
FR6c	As a user, I want inconsistent requirements to be flagged, so conflicting statements don't silently coexist.	✓ Two requirements giving different values for the same attribute are flagged as a pair, each showing the other's ID.	X Requirements on unrelated topics are not paired.	E-C Sprint 2
FR6d	As a user, I want directly contradictory requirements to be flagged, so I catch outright conflicts.	✓ An explicit contradiction (for example “must” vs “must not” on the same behaviour) is flagged with both requirement IDs shown.	X Near-duplicate wording that is not a true contradiction is not flagged.	E-C Sprint 2

FR7 — Explain and Resolve

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR7a	As a user, I want a flagged requirement to be visibly highlighted, so I notice it without scanning the whole list.	✓ Flagged requirements show a distinct icon and colour in the requirement list and can be filtered (FR26c).	X A requirement with several flags shows the flag count, not just the first flag.	E-C Sprint 2
FR7b	As a user, I want an explanation of why something was flagged, so I understand the issue without guessing.	✓ Every flag shows a plain-language reason (not just a code) and the text that triggered it.	X A reason is always present, even for automatically detected edge cases; it is never blank.	E-C Sprint 2
FR7c	As a user, I want the system to ask me a clarifying question on a flagged issue, so I can resolve it in context.	✓ A clarification prompt is shown on the flag and accepts my answer; the answer is saved and the requirement is re-analysed.	X An unanswered prompt does not block the workflow; it stays open and generation can still proceed (FR8a).	E-C Sprint 3

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR7d	As a user, I want to manually resolve a flagged issue myself, so I'm not forced to answer a system-generated question.	✓ "Mark resolved" closes the flag and records who resolved it and when.	X Resolving a flag does not edit the requirement text; editing text and resolving are separate actions. X A resolved flag can be reopened.	E-C Sprint 3
FR7e	As a user, I want a suggested fix where possible, so I can accept it instead of writing one myself.	✓ A suggested fix is shown as replacement text; accepting it edits the requirement, logs the change (FR22a), and re-runs validation.	X When no suggestion is available the interface says "No suggestion available", not a placeholder.	E-C Sprint 3

FR8 — Generate Despite Unresolved Issues

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR8a	As a user, I want to generate diagrams even if some issues are unresolved, as long as the warnings stay visible, so I'm not blocked by minor ambiguities.	✓ When open flags exist, a "Generate anyway" option is offered and generation proceeds. ✓ All flag types are non-blocking; there is no blocking-severity tier.	X Generation with open flags never hides them; the warnings carry into FR8b.	E-C Sprint 3
FR8b	As a user, I want unresolved warnings to remain visible on the generated diagram, so I don't forget about them.	✓ Elements derived from a requirement with a open flag show a warning indicator that links back to the flag.	X Resolving the flag removes the warning from the diagram immediately; reopening the flag restores it.	E-C Sprint 3

FR9 — Recommend and Confirm Diagram Types

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR9a	As a user, I want to see recommended diagram types with a reason for each, so I understand why they were suggested.	✓ Each recommendation shows a one-line justification tied to the extracted content.	X If the input has too little information for any diagram type, the system shows "not enough information" instead of forcing a recommendation.	E-D Sprint 2
FR9b	As a user, I want to add a diagram type the system didn't recommend, so I'm not limited to its suggestions.	✓ I can add any of the 6 supported types regardless of the recommendation.	X Adding a type with clearly insufficient source data (for example, no interactions for a Sequence diagram) still proceeds, with a assumption warning (DR2).	E-D Sprint 3
FR9c	As a user, I want to remove a recommended diagram type I don't need, so I only generate what's useful.	✓ Deselecting a type removes it from the generation queue.	X Deselecting all types disables Confirm and shows a message; nothing is generated silently.	E-D Sprint 3
FR9d	As a user, I want to confirm my final diagram-type selection, so generation only starts on what I've approved.	✓ Confirmation starts the generation run for the selected types; the selection is locked for the duration of that run.	X Confirming with zero types selected is blocked. X After the run, I can generate additional types later; each is a new run with its own confirmation.	E-D Sprint 1

FR10 — Generate Diagrams + Progress

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR10a	As a user, I want to generate a Use Case diagram, so I can visualize actor–system interactions.	✓ A diagram is produced from the confirmed requirements; actors and use cases match the extraction.	X Insufficient actor/use-case data produces a partial diagram with assumption markers (DR2), not a failure.	E-D Sprint 1
FR10b	As a user, I want to generate a Class diagram, so I can visualize entities and their relationships.	✓ Classes and relationships match the extracted entities and relationships.	X An entity with no relationships still appears as a isolated class.	E-D Sprint 2
FR10c	As a user, I want to generate a Activity diagram, so I can visualize process flow.	✓ The flow matches the extracted processes in a coherent order.	X Ambiguous ordering (flagged in FR4e) is shown as a branch or assumption, not as a arbitrary guess presented as fact.	E-D Sprint 2
FR10d	As a user, I want to generate a Sequence diagram, so I can visualize interaction order.	✓ Lifelines and messages match the extracted interactions in inferred order.	X A missing interaction order falls back to the “order uncertain” marking from FR4e.	E-D Sprint 3
FR10e	As a user, I want to generate a DFD, so I can visualize data movement.	✓ Processes, data flows, data stores, and external entities are categorized per DR1e.	X A data flow with no clear source or target is shown but visually flagged as incomplete.	E-D Sprint 3
FR10f	As a user, I want to generate a Component diagram, so I can visualize system structure.	✓ Components and interfaces are derived from the grouping of processes and entities.	X Insufficient structural information produces a coarse-grained diagram with a assumption note, not a fabricated fine-grained one.	E-D Sprint 4
FR10g	As a user, I want to see a progress/status indicator during generation, so I know the system is working.	✓ A visible indicator shows the current stage (for example “Generating diagram 2 of 3”) for the whole run. ✓ Start and end timestamps of every run are recorded (feeds FR25e and NFR1a).	X A run longer than 120 seconds still shows live progress, not a frozen bar.	E-D Sprint 1

FR11 — Completion Notification

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR11a	As a user, I want a in-app notification when generation finishes, so I know when to return to it.	✓ A notification appears in the notification list whether or not I have left the project, and stays until I read it.	X A generation or regeneration failure also triggers a notification, worded differently from success.	E-D Sprint 2
FR11b	As a user, I want a optional email notification when generation finishes, so I know even if I've closed the tab.	✓ If enabled, a email is sent to my account address on completion or failure.	X Email notifications are off by default; no email is sent unless I turn them on.	E-D Sprint 4
FR11c	As a user, I want to turn email notifications on or off, so I control how I'm notified.	✓ The toggle is in my account settings, persists across sessions, and governs FR11b.	X Turning it off during a run does not cancel a email already queued for that run.	E-D Sprint 4

FR30 — External-Service Failure Handling

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR30a	As a user, I want my request to wait and retry automatically when the AI service is rate-limited, so a temporary limit doesn't lose my work.	✓ On a rate-limit or quota response the request is queued, the status shows "Queued — waiting for the AI service", and it is retried automatically with increasing delays (up to 3 retries within 5 minutes).	X When retries are exhausted the run moves to the failed state defined in FR30b. X Queued requests from different users are served in first-come, first-served order.	E-D Sprint 2
FR30b	As a user, I want a clear message and a Retry option when generation or extraction fails, so I know what happened and can try again without redoing my work.	✓ A plain-language failure message is shown (no raw error text) with a Retry button. ✓ My inputs, confirmed extraction, and diagram edits are not lost. ✓ The technical error is written to the error log (FR25d).	X A failure in one diagram of a multi-diagram run does not discard the diagrams that succeeded.	E-D Sprint 2
FR30c	As a user, I want to be told when the server is waking up, so a slow first request isn't mistaken for an error.	✓ If the backend has not responded within 3 seconds, the interface shows "Server is starting, this can take up to 60 seconds" and retries automatically.	X After 90 seconds without a response the interface shows an error with a Retry button.	E-D Sprint 1

FR12 — Editable Diagram Draft

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR12a	As a user, I want to add an element to a diagram, so I can include something the system missed.	✓ The new element appears in the editor and is persisted when I press Save (each Save creates a new version, FR14d).	X Adding an element with the same name as an existing one in that diagram is flagged before saving, not silently allowed.	E-E Sprint 2
FR12b	As a user, I want to remove an element from a diagram, so I can delete something incorrect.	✓ The removed element is gone after Save.	X Removing an element that has relationships shows a warning listing the relationships that will be removed.	E-E Sprint 2
FR12c	As a user, I want to rename an element, so labels match my intended terminology.	✓ The new label appears everywhere the element appears in that diagram.	X Renaming does not change the same element in other diagrams unless I choose "Apply to all diagrams" (see FR17e).	E-E Sprint 2
FR12d	As a user, I want to move/reposition an element, so I can improve the diagram's layout.	✓ The element's position updates and persists after Save and reload.	X Relationship lines re-route automatically and are not broken by moving.	E-E Sprint 2
FR12e	As a user, I want to change a relationship between elements, so I can correct how they connect.	✓ The relationship type and direction update and persist after Save.	X A relationship type that is invalid for that diagram's notation (per DR1) is rejected with an explanation.	E-E Sprint 2

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR12f	As a user, I want to edit diagram-specific properties (for example multiplicity or stereotype), so the diagram is technically accurate.	✓ Property edits save and render correctly.	X An invalid value for that diagram type (for example, non-numeric multiplicity) is rejected inline with the accepted format shown.	E-E Sprint 2

FR13 — Regenerate Portion or Whole

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR13a	As a user, I want to select a portion of a diagram and regenerate just that part, so I don't lose unrelated edits.	✓ The selected elements are regenerated; the rest of the diagram is unchanged.	X If the selection has dependencies outside it, a warning lists them before regeneration. X Manual edits inside the selection are replaced only after I confirm.	E-E Sprint 4
FR13b	As a user, I want to regenerate a entire diagram, so I can start that diagram fresh.	✓ The full diagram is regenerated from the current requirements model.	X Before regeneration the current state is saved as a version and a confirmation prompt explains that manual edits will be replaced; they can be recovered by reverting (FR14e).	E-E Sprint 3

FR14 — Modify Requirements, Regenerate, Version History

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR14a	As a user, I want to modify a source requirement, so I can correct or update it.	✓ The edited requirement text is saved with a timestamp and my name, and logged (FR22a). ✓ The statement is re-validated (FR6) and re-extracted; items extracted only from it are updated.	X Items I added or edited manually ("user-added", "user-edited") are never overwritten; if the new text conflicts with one, that item is flagged for review. X Editing a currently flagged requirement re-runs validation so no stale flag remains.	E-E Sprint 3
FR14b	As a user, I want the system to tell me which diagrams are affected by a requirement change, so I know what needs attention.	✓ An affected-diagrams list is shown immediately after a requirement edit.	X If no diagram is linked to the requirement, "No diagrams affected" is shown, not a blank or error state.	E-E Sprint 3
FR14c	As a user, I want affected diagrams to be regenerated after I approve a requirement change, so they stay accurate.	✓ When I approve propagation (FR17c), the affected diagrams are regenerated, incrementally where possible (NFR6). Nothing is regenerated without my approval.	X A regeneration failure on one diagram does not block the others in the same batch; failures are reported per diagram (FR30b).	E-E Sprint 3
FR14d	As a user, I want to view the version history of a diagram, so I can see how it has changed over time.	✓ The version list shows the latest 20 versions with timestamp, author, and reason (generated, saved edit, regenerated, reverted).	X With only one version, the list shows that version and the message "No earlier versions", not a error. X When a 21st version is created the oldest is dropped.	E-E Sprint 3

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR14e	As a user, I want to revert a diagram to a previous version, so I can undo a unwanted change.	✓ Reverting makes the selected version current; the state before reverting is saved as a new version first (non-destructive).	X Reverting does not delete newer versions.	E-E Sprint 3

FR15 — Visual Customisation Before Export

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR15a	As a user, I want to change a diagram's colour scheme, so it matches my documentation style.	✓ The colour change applies and persists for that diagram. ✓ Structural elements and relationships are unchanged.	X A colour pair with contrast below the WCAG AA ratio (4.5:1 for text) shows a warning before applying.	E-E Sprint 4
FR15b	As a user, I want to change a diagram's layout direction, so it fits my page or document layout.	✓ Direction (top-down or left-right) updates and elements re-flow. ✓ Structural elements and relationships are unchanged.	X If manual positions (FR12d) would be lost by re-layout, a warning is shown first and I can cancel.	E-E Sprint 4
FR15c	As a user, I want to change font size, so the diagram is readable at my intended output size.	✓ Font size (8–24 pt) applies and persists. ✓ Structural elements and relationships are unchanged.	X A size that causes label overlap or clipping is handled by auto-wrapping or resizing the box, not silently broken.	E-E Sprint 4

FR29 — Generated-Diagram Validation

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR29a	As a user, I want each generated diagram to be validated automatically, so errors are caught before I rely on it.	✓ After generation, each diagram is checked against its notation rules (DR1), against the confirmed extracted items (for example, a extracted actor missing from a Use Case diagram), and for structural problems (for example, a Class diagram relationship with no target).	X A diagram with no issues shows “No validation issues”, not a blank state.	E-E Sprint 3
FR29b	As a user, I want validation warnings and suggestions to be shown on the diagram, so I can decide what to change.	✓ Each warning is anchored to the affected element (or the diagram) with a plain-language description and, where possible, a suggested fix I can apply in one click.	X Warnings persist until the cause is fixed; they survive reload and are recorded in export metadata (FR18d).	E-E Sprint 3
FR29c	As a user, I want to either edit the diagram or accept it as it is, so I stay in control of the outcome.	✓ “Accept as-is” marks a warning as accepted with my name and date; “Edit” opens the editor on the affected element.	X Accepted warnings stay visible in a muted style and re-appear if the diagram is regenerated and the issue returns.	E-E Sprint 4

FR16 — Shared Requirements Model

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR16a	As a user, I want all diagrams in a project to be generated from one shared requirements model, so the same actor, entity, or process is never described two different ways.	✓ Every diagram element stores a reference to a model item ID (requirement statement or confirmed extracted item), not its own copy of the data.	X Test: renaming a actor in the model updates the reference seen by every diagram that uses it (verified by automated test).	E-F Sprint 1
FR16b	As a user, I want a change to the shared requirements model to be reflected in every dependent diagram's traceability view, so traceability stays current.	✓ The traceability view of every dependent diagram updates automatically after a model change.	X Removing a requirement entirely updates all dependent traceability links; no dangling references remain.	E-F Sprint 3

FR17 — Traceability & Consistency

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR17a	As a user, I want to click a diagram element and see which requirement(s) it came from, so I can verify its origin.	✓ Selecting a element shows all linked source requirements (IDs, text, document location), even when there are several.	X An element with no source requirement shows “No source requirement — assumption”, not a blank or broken link.	E-F Sprint 2
FR17b	As a user, I want to see a list of requirements not represented in any diagram, so nothing gets silently dropped.	✓ An “Unrepresented requirements” list is available and matches the FR20c count exactly.	X A requirement that becomes represented after a new generation leaves the list automatically.	E-F Sprint 3
FR17c	As a user, I want to be prompted to propagate a requirement change to affected diagrams, so I don't have to remember to update each one.	✓ After a requirement edit that affects existing diagrams, a prompt offers “Regenerate now” or “Later”.	X Choosing Later leaves each affected diagram marked “out of date”; it is never silently marked up to date.	E-F Sprint 3
FR17d	As a user, I want to be warned when a manual diagram edit drifts from the current requirements, so I know my diagram and my requirements have diverged.	✓ A drift warning appears when a saved manual edit contradicts its linked requirement (for example, removing a element the requirement mandates, or adding a relationship it does not support).	X The warning clears automatically once the requirement is updated to match the diagram or the edit is reverted.	E-F Sprint 3
FR17e	As a user, I want to be flagged when the same element is named inconsistently across diagrams, so my diagram set stays coherent.	✓ A naming-mismatch flag appears when the same model item has different labels in different diagrams of one project, with a “Apply this name to all diagrams” action.	X Two different items that share a name are not flagged; matching is by model item ID, not by text.	E-F Sprint 4

FR18 — Export Formats

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR18a	As a user, I want to export a diagram as PNG, so I can embed it as a image.	✓ A PNG downloads with correct rendering at 2× scale.	X A very large diagram is scaled to at most 8,000 px on its longest side, not silently truncated.	E-G Sprint 1

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR18b	As a user, I want to export a diagram as SVG, so I have a scalable vector version.	✓ An SVG downloads, renders correctly, and scales without quality loss.	X Fonts and colours from FR15 styling are embedded or degrade gracefully, not lost.	E-G Sprint 2
FR18c	As a user, I want to export a diagram as PDF, so I can share a print-ready version.	✓ A PDF downloads on A4 pages with readable labels.	X A diagram larger than one page is scaled to fit one page (landscape when wider than tall); elements are never cut mid-element.	E-G Sprint 2
FR18d	As a user, I want to export a diagram in a editable XML format, so I can continue editing it in another tool.	✓ An XML file in draw.io format (.drawio) downloads and opens correctly in diagrams.net.	X A diagram with unresolved warnings (FR8b) still exports; warnings are recorded as metadata, not blocked.	E-G Sprint 3
FR18e	As a user, I want to export a diagram as PlantUML text, so I can version-control it as text.	✓ A .puml file downloads containing valid PlantUML that re-renders to a equivalent diagram.	X Special characters in labels are escaped so they do not break the syntax. X DFDs use PlantUML rectangle, ellipse, and database shapes for external entity, process, and data store, with a header comment naming the mapping.	E-G Sprint 3
FR18f	As a user, I want a one-click export option, so I don't have to configure format or settings every time.	✓ One action exports in my default format.	X If I have not set a default, PNG is used; the action does not error. X The default format can be changed in account settings.	E-G Sprint 4

FR19 — Pipeline Guidance & Contextual Help

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR19a	As a new user, I want step-by-step guidance through each pipeline stage, so I know what to do next.	✓ Guided prompts appear at each stage from input to export.	X I can dismiss the guidance permanently; it does not reappear in later sessions unless I re-enable it in settings.	E-H Sprint 4
FR19b	As a user, I want a contextual explanation when I don't recognize a diagram notation element, so I can learn as I go.	✓ Hovering or clicking a element shows a plain-language explanation of its notation.	X Explanations exist for every notation element used in all 6 diagram types, not a subset.	E-H Sprint 4

FR20 — Coverage Summary

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR20a	As a Project Manager (a project Owner), I want to see how many requirements have generated diagrams, so I can track modelling progress.	✓ The count is shown as “n of N requirements” and updates as diagrams are generated.	X Deleting the only diagram that covers a requirement decrements the count.	E-I Sprint 4

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR20b	As a Project Manager (a project Owner), I want to see how many requirements have unresolved ambiguity flags, so I know what still needs attention.	✓ The count of requirements with at least one open flag is shown and updates as flags are resolved.	X A newly added requirement that is flagged immediately is counted right away.	E-I Sprint 4
FR20c	As a Project Manager (a project Owner), I want to see how many requirements aren't represented in any diagram, so I can catch coverage gaps.	✓ The count matches the FR17b list exactly.	X The summary count and the detailed list can never disagree.	E-I Sprint 4

FR21 — Combined Report Export

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR21a	As a Project Manager (a project Owner), I want the system to compile requirements, diagrams, and open issues into one report, so I don't have to assemble it manually.	✓ The report contains the current requirement statements, the latest version of each diagram, and all open flags and warnings as of generation time.	X With zero diagrams the report is still produced, with a note that no diagrams exist yet.	E-I Sprint 4
FR21b	As a Project Manager (a project Owner), I want to download that report as a single PDF, so I can share it outside the tool.	✓ The report downloads as one PDF file.	X A project at the limits (15 documents / 100 pages) still produces a report within 60 seconds.	E-I Sprint 4

FR22 — Activity Log

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR22a	As a Project Manager (a project Owner), I want to see who changed a requirement and when, so I can track requirement history.	✓ Every requirement edit is logged with user, timestamp, and old/new text; only Owners (and administrators, once they hold a role on the project) can view the log.	X A bulk change (for example, a re-upload) is logged per requirement, not as one opaque entry.	E-I Sprint 3
FR22b	As a Project Manager (a project Owner), I want to see who edited or regenerated a diagram and when, so I can track diagram history.	✓ Every diagram save, regeneration, and revert is logged with user and timestamp.	X A regeneration that the user approved after a requirement change is labelled "regenerated after requirement change", distinguishable from a manual regeneration.	E-I Sprint 3

FR28 — Accounts & Sessions

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR28a	As a new user, I want to sign up with my email and a password, so I can start using the tool.	✓ I enter name, email, and a password of at least 8 characters containing a letter and a digit; a verification email is sent (link valid 24 hours); I can log in only after verifying. ✓ Passwords are stored only as salted hashes.	- X A duplicate email is rejected with a message. - X A weak password is rejected inline with the rule shown. - X An expired link offers to send a new one.	E-A Sprint 1
FR28b	As a new user, I want to sign in with my Google account, so I don't need another password.	✓ "Sign in with Google" creates a account on first use with the Google-verified email and name, and logs me in.	- X If a account with the same email already exists, Google sign-in opens that same account (Google verifies the email); no duplicate account is created. - X If I cancel on Google's screen, I return to the login page with no error state.	E-A Sprint 2
FR28c	As a user, I want to log in with my email and password, so I can access my projects.	✓ Correct credentials on a verified account start a session that expires after 24 hours of inactivity.	- X After 5 failed attempts the account is locked for 15 minutes, and the message never reveals whether the email or the password was wrong. - X Logging in with a unverified email offers to resend the verification email.	E-A Sprint 1
FR28d	As a user, I want to log out, so my session ends on a shared computer.	✓ Logging out invalidates the session on the server immediately.	X Pressing the browser's Back button after logout does not show protected data; it redirects to login.	E-A Sprint 1
FR28e	As a user, I want to reset a forgotten password, so I can regain access to my account.	✓ A reset link is emailed, is single-use, and is valid for 1 hour; setting a new password ends all my other sessions.	- X An unknown email shows the same confirmation message as a known one (no account enumeration). - X For a Google-only account the email tells the user to use Google sign-in.	E-A Sprint 2

FR23 — Admin Account Management

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR23a	As a System Administrator, I want to view a list of all user accounts, so I can see who has access.	✓ The admin dashboard lists all accounts with name, email, sign-up method, and status (active / deactivated).	X The list is paginated (25 per page) and searchable by name or email.	E-J Sprint 3
FR23b	As a System Administrator, I want to add a new user account, so I can onboard someone manually.	✓ The new account receives a invitation email to set a password (or sign in with Google) and can then log in.	- X A duplicate email is rejected with a clear message. - X The invitation link expires after 7 days and can be re-sent.	E-J Sprint 3

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR23c	As a System Administrator, I want to deactivate a user account, so I can revoke access when needed.	✓ A deactivated account can no longer log in and its active sessions end immediately.	X Deactivation does not delete the user's projects; they stay accessible to collaborators. X If the user is the sole Owner of a project that has other members, the admin is prompted to reassign ownership to a member. X An administrator cannot deactivate themselves or the last active administrator.	E-J Sprint 3
FR23d	As a System Administrator, I want to reactivate a deactivated account, so I can restore access when it was revoked by mistake or is needed again.	✓ A reactivated account can log in again with its previous data intact.	X Reactivation is not possible after the account's data has been purged (NFR5b, NFR5c).	E-J Sprint 4

FR24 — Collaborator Invitations, Roles & Concurrency

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR24a	As a project owner or System Administrator, I want to invite a collaborator to a project, so we can work together.	✓ I enter a email and choose a role (default Viewer); the invitee receives a email link valid for 7 days; on acceptance the project appears in their list marked "Shared".	X An invitee without a account is taken through sign-up (FR28a / FR28b) and then given access, not left with a failed link. X When a administrator invites, every Owner of the project is notified and the action is logged (FR22).	E-J Sprint 3
FR24bcd	As a project owner, I want to assign a collaborator a role — Viewer, Editor, or Owner, so I can control what each person can see or change (merged from FR24b, FR24c, FR24d).	✓ Viewer: can open the project, view diagrams and requirements, export, and report issues; cannot edit. ✓ Editor: everything a Viewer can, plus edit and generate requirements, extracted items, and diagrams and upload documents; cannot manage roles or delete the project. ✓ Owner: full permissions, including roles, rename, delete, coverage summary, report, and activity log.	X Assigning a role to someone who is not a accepted or pending collaborator is rejected. X The last Owner of a project cannot be demoted.	E-J Sprint 3
FR24e	As a collaborator, I want my assigned role to actually restrict or allow my actions, so permissions are enforced, not just cosmetic.	✓ An action outside my role (for example, a Viewer trying to edit) is rejected on the server with a "not permitted" response, not just hidden in the UI.	X A role downgrade or removal takes effect on my next request, including for a session already open.	E-J Sprint 3
FR24f	As a editor, I want to be warned when someone else changed a requirement or diagram since I opened it, so my save doesn't silently overwrite their work.	✓ Every editable item carries a revision number; a save based on a older revision is rejected with "Changed by <name> at <time>" and the options Reload latest or Save mine as a new version.	X Two people opening the same diagram read-only never conflict; only conflicting saves are affected.	E-J Sprint 3

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR24g	As a project owner, I want to remove a collaborator from my project, so I can revoke access I no longer want to give.	✓ The collaborator loses access immediately and the project disappears from their list; collaborators may also leave a project themselves.	X The last Owner cannot be removed and cannot leave. X The removed user's past edits and log entries stay in the project.	E-J Sprint 4

FR25 — Admin System Monitoring

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR25a	As a System Administrator, I want to see the active user count, so I can gauge current load.	✓ The dashboard shows the number of accounts with a authenticated request in the last 15 minutes, refreshed at least every minute.	X The count reflects recent activity, not total registered users.	E-J Sprint 4
FR25b	As a System Administrator, I want to see the total number of projects in the system, so I can gauge overall usage.	✓ The dashboard shows the total project count.	X Deleted projects (FR3e) are excluded.	E-J Sprint 4
FR25c	As a System Administrator, I want to see storage used against defined limits, so I can monitor capacity.	✓ The dashboard shows system-wide storage used against the configured storage cap, and reminds me of the per-project limits (10 MB per file, 15 documents, 100 pages).	X Usage at or above 80% of the cap is visually flagged, not just shown as a raw number.	E-J Sprint 4
FR25d	As a System Administrator, I want to see recent error and failure logs, so I can diagnose problems.	✓ The dashboard lists the most recent errors with timestamp, type, and severity; log entries never contain document content.	X The log is filterable by error type and severity, not one undifferentiated stream.	E-J Sprint 4
FR25e	As a System Administrator, I want to see generation-time statistics against the NFR1 target, so I can monitor performance health.	✓ The dashboard shows median and 95th-percentile generation time versus the NFR1 target (median ≤ 60 s, 95th percentile ≤ 120 s).	X Runs longer than 120 seconds are listed individually, not buried in the average. X With no runs yet, the panel shows “No generations recorded yet”, not a error.	E-J Sprint 4
FR25f	As a System Administrator, I want to view the system configuration, so I can confirm the limits and settings the system is running with.	✓ A read-only page shows file-size, document, and page limits, retention period, operating hours, storage cap, AI provider name and model, and notification settings.	X Secrets (API keys, credentials) are never displayed, not even masked partially.	E-J Sprint 4

FR26 — Search and Filter

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR26a	As a user, I want to search requirements and diagrams by keyword, so I can find relevant content quickly.	✓ Keyword search inside the current project matches requirement text, extracted item names, and diagram titles and element labels, and lists the matches.	X No match shows “No results”, not a blank or broken state.	E-K Sprint 4
FR26b	As a user, I want to search by requirement or diagram ID, so I can jump directly to a known item.	✓ An exact ID (for example R-014) opens that item directly.	X A non-existent ID shows a clear “not found” message.	E-K Sprint 4
FR26c	As a user, I want to filter by status, so I can focus on items needing action.	✓ Status filters: Flagged (a flag was ever raised), Unresolved (at least one flag currently open), Exported (diagram exported at least once).	X Filters from different categories (status + diagram type) combine with AND; values within one category combine with OR.	E-K Sprint 4
FR26d	As a user, I want to filter by diagram type, so I can view only the diagrams I care about.	✓ The list narrows to the selected diagram type(s).	X A type with no diagrams shows a empty state, not a error.	E-K Sprint 4

FR27 — In-Tool Feedback

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
FR27a	As a user, I want to report a issue on a specific diagram, so I can flag a problem without leaving the app.	✓ A feedback form is available on every diagram and records the diagram ID and version, a category (inaccuracy, bug, suggestion), and a description.	X Submitting without a description of at least 10 characters is rejected inline.	E-H Sprint 4
FR27b	As a user, I want to report a issue or feedback on a specific requirement, so I can flag a problem tied to that item.	✓ A feedback form is available on every requirement and records its ID, a category, and a description.	X Feedback on a requirement that is later deleted is retained (shown as a orphaned entry) until the project is deleted.	E-H Sprint 4
FR27c	As a project owner or System Administrator, I want to view the feedback reports submitted, so I can act on problems users found.	✓ Owners see feedback for their projects; administrators see all feedback; each report shows author, target, category, description, and status (open / closed) and can be closed.	X Viewers and Editors who are not Owners cannot see other people's reports, only their own.	E-H Sprint 4

2. Non-Functional Requirement Stories

NFR1 — Performance

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR1a	As a user, I want diagram generation to complete quickly for a typical document, so I'm not left waiting indefinitely.	✓ On the team's benchmark set of typical documents (up to 20 pages / 8,000 words, up to 3 selected diagram types), median generation time is at most 60 seconds and the 95th percentile at most 120 seconds.	X Any run longer than 120 seconds is recorded and shown to administrators (FR25e); it is never accepted silently.	E-L Sprint 3

NFR2 — Usability

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR2a	As a new user, I want to complete upload → generate → export unaided within 1 hour, so I don't need formal training.	✓ In a usability test with at least 5 first-time users who are not on the project team, at least 4 complete the task unaided within 1 hour.	X Every step where two or more testers get stuck is logged as a usability defect and prioritised, not dismissed.	E-L Sprint 4

NFR3 — Security — Data Protection

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR3a	As a user, I want my uploaded documents and extracted data to be encrypted at rest, so they are protected if storage is compromised.	✓ Stored files and database content are encrypted at rest (AES-256 via the storage/database provider); verified by a checklist against the provider configuration.	X Keys are managed by the platform provider; a documented key-rotation procedure exists; custom key management is out of scope for this course project.	E-A Sprint 2
NFR3b	As a user, I want my data to be encrypted in transit, so it can't be intercepted while uploading or downloading.	✓ All traffic uses HTTPS/TLS; plain HTTP requests are redirected to HTTPS.	X A pre-release audit checks every endpoint and finds none served over plain HTTP.	E-A Sprint 1

NFR4 — Authentication & Access Control

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR4a	As a user, I want to log in before accessing any project, so my data isn't publicly reachable.	✓ Unauthenticated page requests redirect to login; unauthenticated API requests return 401.	X A request with a expired or invalid session token is rejected the same way, on the server.	E-A Sprint 1
NFR4b	As a user, I want access control enforced per project, so only people I've granted access can view or edit it.	✓ A user who is not a member of a project cannot open it or any of its documents, diagrams, or files.	X Guessing a project, document, or diagram ID in a URL or API call returns "not found" (not "forbidden"), so the resource's existence is not revealed.	E-A Sprint 2

NFR5 — Transparency & Retention

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR5a	As a user, I want to view the Privacy Policy from within the app, so I understand how my data is handled.	✓ The policy is linked from the footer and the consent screen and states: what is collected; that content is sent to a external AI provider; storage and encryption; the 30-day inactivity retention; deletion on request; administrator access; and that anonymised, content-free audit records are kept for 90 days.	X The policy shows a version number and effective date; the change history is kept.	E-A Sprint 1
NFR5b	As a user, I want my data to be deleted automatically after 30 days of account inactivity, so old data doesn't linger indefinitely.	✓ An account is inactive after 30 consecutive days with no login or authenticated activity. A warning email is sent once between day 25 and day 30. At day 30 the account, its documents, diagrams, versions, and feedback are permanently deleted.	X Projects the account solely owns that still have Editors active in the last 30 days are transferred to the most recently active Editor instead of being deleted; other sole-owned projects are deleted. X Projects with other Owners stay; the purged account is removed from them. X If the warning email cannot be sent, the purge still runs and the failure is logged.	E-A Sprint 3
NFR5c	As a user, I want to request immediate deletion of my data, so I control removal without waiting 30 days.	✓ "Delete my data" purges my account and all data I solely own immediately after I confirm (typing DELETE, and re-entering my password if I use one).	X Before confirming, I see which shared projects will be deleted and how many people they affect; those people are notified. X Projects with other Owners stay; I am removed from them. X Only anonymised, content-free audit records (event type, timestamp, hashed user ID) remain, for 90 days.	E-A Sprint 3
NFR5d	As a user, I want to be told and asked for consent before my content is sent to a external AI provider, so I know how my requirements are processed.	✓ Before my first submission a consent screen states that text is sent to the external AI provider (currently Google Gemini) and links to the Privacy Policy; my consent is recorded with timestamp and policy version. ✓ Without consent, no content is submitted for AI processing.	X If the AI-related section of the policy changes, I must consent again before the next submission. X Withdrawing consent stops further AI processing; my stored data remains until deleted.	E-A Sprint 1

NFR6 — Incremental Regeneration

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR6a	As a user, I want incremental regeneration to update only the affected diagram parts, so unrelated parts aren't disturbed.	✓ After regeneration, elements outside the affected scope are unchanged in label, properties, and position.	X A change with cascading effects (for example, a renamed actor used everywhere) correctly includes every dependent element in the scope.	E-L Sprint 4

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR6b	As a user, I want incremental regeneration to complete quickly, so small changes feel fast.	✓ A change to a single requirement statement (affecting at most 10 diagram elements) is regenerated within 30 seconds.	X A change that cannot meet 30 seconds shows live progress (FR10g), not a hung screen.	E-L Sprint 4

NFR7 — Availability

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR7a	As a user, I want the system to be available during weekday operating hours, so I can rely on it during my working week.	✓ Operating hours are Monday–Friday, 08:00–20:00 IST. Monthly availability during those hours is at least 95%, measured by an external uptime monitor that calls the health route every 10 minutes.	X An unplanned outage during operating hours is recorded as an incident with start and end time in the error log (FR25d).	E-L Sprint 2
NFR7b	As a System Administrator, I want to schedule maintenance downtime on weekends, so upkeep doesn't disrupt weekday use.	✓ Maintenance windows can be scheduled only on Saturday or Sunday; users see a banner at least 24 hours before.	X A window that overlaps a weekday is rejected with an explanation.	E-L Sprint 4

NFR8 — Browser Compatibility

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR8	As a user, I want the system to work correctly on the latest two versions of Chrome, Firefox, and Edge, so I can use my browser of choice (merged from NFR8a, NFR8b, NFR8c).	✓ The full functional test suite passes on desktop Chrome, Firefox, and Edge, each on its latest two stable major versions at the time of testing, at window widths of 1280 px and above.	X A browser-specific rendering or behaviour bug is treated as a defect against this story, not an untracked issue. X Mobile and tablet layouts are out of scope.	E-L Sprint 4

NFR9 — Scalability

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
NFR9a	As a product owner, I want the system to support at least 5 concurrent users without breaching the NFR1 target, so small teams can use it together reliably.	✓ A load test with 5 users generating simultaneously still meets NFR1.	X A 6th simultaneous generation request is queued (status “Queued”) and served in order; it does not cause failures for the other 5.	E-L Sprint 4

3. Domain Requirement Stories

DR1 — Notation Conventions

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
DR1a	As a user, I want Use Case diagrams to follow standard UML conventions, so anyone familiar with UML interprets them correctly.	✓ Actors, use cases, system boundary, and relationships render per UML notation.	X Include and extend relationships render with correct dashed-arrow notation and stereotypes, not generic lines.	E-D Sprint 1
DR1b	As a user, I want Class diagrams to follow standard UML conventions, so class structure is unambiguous.	✓ Classes (name, attributes, methods) and relationships render per UML notation.	X Multiplicity (1, 0..*, and so on) renders correctly where it can be extracted.	E-D Sprint 2
DR1c	As a user, I want Activity diagrams to follow standard UML conventions, so process flow is unambiguous.	✓ Initial/final nodes, actions, and flows render per UML notation.	X Decision and merge points render as diamonds with guard labels, not just arrows.	E-D Sprint 2
DR1d	As a user, I want Sequence diagrams to follow standard UML conventions, so interaction order is unambiguous.	✓ Lifelines, activation bars, and messages render per UML notation.	X Return messages (where inferable) are dashed and distinct from calls.	E-D Sprint 3
DR1e	As a user, I want DFDs to follow standard DFD conventions, so processes, data flows, data stores, and external entities are correctly distinguished.	✓ The Yourdon/DeMarco-style symbol set is used: circle/rounded shape for process, open-ended rectangle for data store, square for external entity, labelled arrow for data flow.	X The four element kinds are visually distinct symbol types, never interchangeable boxes.	E-D Sprint 3
DR1f	As a user, I want Component diagrams to follow standard UML conventions, so system structure is unambiguous.	✓ Components, provided and required interfaces, ports, and dependencies render per UML notation.	X A dependency with no identifiable interface renders as a dependency arrow, not as an interface.	E-D Sprint 4

DR2 — Assumptions vs. Stated Facts

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
DR2a	As a user, I want elements with insufficient requirement backing to be marked “assumption”, so I know what the system inferred rather than what was stated.	✓ Any diagram element with no linked requirement statement is tagged “assumption”.	X The tag is not lost after a edit unless the element is linked to a requirement (the gap is actually resolved).	E-D Sprint 1
DR2b	As a user, I want assumption-marked elements to look different from requirement-derived ones, so I can prioritise reviewing the uncertain ones.	✓ Assumption elements use a distinct icon and dashed outline in the editor.	X The distinction is preserved in every export: PNG/SVG/PDF show the visual style; the XML and PlantUML exports carry a “assumption” marker (stereotype or comment).	E-D Sprint 3

4. Technical Enabler Stories

TE — Technical Enablers

ID	User Story (front of card)	Success Criteria	Failure / Edge Criteria	EPIC · Sprint
TE1	As a development team member, I want a GitHub repository with a pull-request workflow and continuous integration, so all work is integrated safely and every contribution is attributable.	✓ All 10 members are contributors; the main branch is protected and needs one approving review; CI runs lint and tests on every pull request; each member commits under their own account. ✓ Project communication runs on the team's Slack workspace, including CI-failure notifications.	- X Direct pushes to main are blocked. - X Commit authorship is checked at each sprint review.	E-M Sprint 1
TE2	As a development team member, I want a automated deployment pipeline with a health route and uptime monitoring, so the app is always deployed and the free-tier backend doesn't stay asleep.	✓ A merge to main deploys the frontend to Vercel and the backend to Render; the backend exposes a health route; a external uptime monitor calls it every 10 minutes.	- X Secrets live only in platform environment settings, never in the repository. - X A failed deployment is visible to the team, and the previous deployment can be restored.	E-M Sprint 1
TE3	As a development team member, I want a database and file storage that persist independently of the backend server, so user data survives restarts and redeployments.	✓ Data and uploaded files remain available after the backend restarts or is redeployed; database changes are applied through versioned migrations.	X Storage use is measured so that FR25c can report it.	E-M Sprint 1
TE4	As a development team member, I want one AI-service module that all AI calls go through, so the provider and model can be changed without touching feature code.	✓ The module wraps Google Gemini Flash; provider, model, and API keys are configuration; keys are never sent to the browser; logs contain metadata only, never document content. ✓ It exposes hooks for rate-limit queuing and retry (FR30a) and failure reporting (FR30b).	X If more than one API key is configured for failover, use is checked against the provider's current terms before it is enabled.	E-M Sprint 1
TE5	As a development team member, I want a email-sending service integrated into the backend, so verification, reset, invitation, and notification emails can be sent.	✓ Emails send from one configured sender address through a transactional email provider.	X A send failure is logged and retried once; it never blocks the main user action.	E-M Sprint 1
TE6	As a development team member, I want a scheduled-jobs runner, so time-based rules (retention purge, warnings, invitation expiry) run reliably.	✓ A daily job handles the day-25 warning, the day-30 purge, and expiry of invitations and verification links; each run is logged and safe to repeat.	X If a run is missed (for example, the server was asleep), the next run processes everything now due, based on stored timestamps.	E-M Sprint 3